PRODUCT REQUIREMENTS DOCUMENT · HARSHIT DEV YADAV

Duolingo — Notifications

Durable habit formation without burning the notification channel.

Author Harshit Dev Yadav **Product** Duolingo (Consumer / EdTech) **Status** Sample PRD

North-star D30 streak-maintenance rate

⚡ TL;DR

Product: Duolingo notification / re-engagement system.

Who: Streak holders at risk of losing their streak (D3-D30 users). Duolingo has 575M+ total users, 42M MAU. The retention engine runs on streaks.

Why now: Notification fatigue is driving opt-outs. Up to 3 push notifications per day for at-risk users. Open rate declining QoQ. When a user opts out, the channel is gone — permanently.

Launch: Phased, with frequency-cap guardrail hard-enforced from day one.

575M+	~42%	+8pp
TOTAL USERS	D30 STREAK BASELINE	TARGET LIFT (60 DAYS)

💡 CORE INSIGHT

More notifications has diminishing returns and a **real downside** — fatigue → opt-out → the channel is lost permanently. When a user opts out, they almost never re-enable in device settings. The goal is **habit retention, not notification volume**. Sending right beats sending more.

Background & Problem

Duolingo's retention engine runs on streaks. Loss aversion kicks in around day 3 and compounds — a 14-day streak is meaningfully more painful to lose than a 1-day streak. But the notification layer that supports streak maintenance is optimising for volume, not habit.

Current state: Up to 3 push notifications/day for at-risk users. Open rate declining quarter-on-quarter. Opt-out rate creeping up. The system also uses passive-aggressive copy ("These reminders don't seem to be working. We'll stop sending them for now.") — a guilt-trigger that accelerates opt-out without proportional retention lift.

The trust account framing: A notification opt-out destroys future retention capacity indefinitely. Each poor send is a permanent withdrawal. Sends matched to habit windows build the balance.

Behavioural Economics at Play

– **Loss aversion (Kahneman):** "You'll lose your 14-day streak" outperforms "Keep your 14-day streak" on streak-maintenance specifically

- **Variable reward (Skinner):** Unpredictable content variation fights fatigue better than a reliable daily ping
- **Implementation intentions (Gollwitzer):** Sends matched to the user's existing practice window increase follow-through vs. batch-job timing

Goals & Non-Goals

Goals

- Personalise send-time to each user's historical practice window (ML-timed)
- Hard frequency cap: max 1 notification/day, 5/week, enforced at send layer
- Streak-loss-aversion copy: message references specific streak length and names the loss
- Content variation pool (min 6 variants per scenario) to fight repetition fatigue
- Lesson-completion suppression: no notification if lesson done that day
- Instrument per-notification opt-out rate and D30 streak-maintenance lift

Non-Goals

- New lesson types or content changes
- Changes to streak-freeze or streak-repair mechanics
- Changes to the lesson engine or scoring
- Paid-tier or Super Duolingo changes
- Win-back campaigns for users who have already uninstalled
- Email or in-app notification redesign (separate surface)

User Stories / JTBD

- *When I'm at risk of losing a streak I care about, I want a reminder that feels personal so I act on it.*
- *When I've already done my lesson today, I don't want to receive a notification.*
- *When I've been nagged three days in a row with the same message, I want to turn off notifications — and then I do.*
- *When I get a passive-aggressive guilt notification, I feel manipulated and consider uninstalling.*

Functional Requirements

P0 — MUST SHIP

— ML send-time personalisation: model each user's practice window (time-of-day + day-of-week); send within ±1h of that window

— Hard frequency cap: max 1/day, 5/week — enforced at send layer (not application layer), zero exceptions

— Streak-aware copy: message references specific streak length ("Your 14-day streak is at risk") — no generic copy

— Lesson-completion suppression: if lesson completed in past 12h, suppress all sends for that day

— Content variation: min 6 copy variants per send scenario; same variant not shown twice consecutively

P1 — SHOULD SHIP

— Loss-framing A/B test: "You'll lose your 14-day streak" vs. "Keep your 14-day streak" — measured on D7 retention, not open rate

— Per-user opt-out rate monitoring dashboard (ops-facing, daily refresh)

— Retire the passive-aggressive notification type entirely

P2 — NICE TO HAVE

— Cross-channel coordination: suppress push if email was opened same-day

— Streak-milestone celebration (7, 30, 100 days) with variable reward (collectible badge teaser)

— Smart re-permission prompt for users who opted out >30 days ago

Acceptance Criteria

REQUIREMENT	PASS / FAIL TEST
Send-time personalisation ≥80% of sends occur within the user's historical ±1h practice window	MEASURABLE FROM SEND LOGS
Frequency cap No user receives >1 notification in any 24h window	ENFORCED AT SEND LAYER, QUERY-VERIFIABLE
Copy specificity 100% of notifications reference the user's actual streak count	ZERO GENERIC-COPY SENDS IN PRODUCTION
Lesson suppression 0 notifications sent to users who completed a lesson in the past 12h	JOIN SEND LOG + LESSON LOG, COUNT VIOLATIONS

Success Metrics

★ NORTH-STAR METRIC

D30 streak-maintenance rate for notified users

Baseline ~42% → Target ~50% (+8pp) in 60 days

Guardrails (must not worsen):
Notification opt-out rate · App uninstall rate (D7) · Per-user frequency (hard cap, zero exceptions)

We are NOT measuring: Open rate · Click-through rate · Total sends

Wireframes / User Flows

FLOW 1 — SEND-TIME PERSONALISATION

```
User completes lesson [timestamp logged] → ML model updates practice window → next-day send queued within ±1h window → lesson-completion check at send time → send or suppress
```

FLOW 2 — LOSS-FRAMING NOTIFICATION

```
Copy: "Your [N]-day streak ends at midnight. 3 minutes to keep it." → tap → lesson opens at next lesson in sequence → completion suppresses all sends for that day
```

FLOW 3 — CONTENT VARIATION ROTATION

```
6+ variants per scenario → variant assignment avoids consecutive repeats → fatigue score tracked per user → copy refresh triggered when all variants exhausted
```

Risks & Open Questions

KNOWN **ML cold-start:** New users have no historical practice window. Default to 8pm local time for first 7 days; personalise after 3+ data points.

KNOWN **Loss vs. gain framing:** Loss framing increases streak maintenance short-term but may increase opt-outs for users who find it manipulative. Do not ship at 100% without A/B data — the test resolves this.

KNOWN **Passive-aggressive notification retirement:** This type has non-zero CTR because guilt works short-term. Removing it will show a short-term CTR dip. Track D30 opt-out rate, not CTR, as the signal.

OPEN **Lesson-completion timestamp granularity:** Does the current data pipeline log lesson completions with sub-1h granularity? Required for same-day suppression. Confirm with data eng before P0 commit.

Rollout Plan

PHASE 1

Days 0-14 · 5% of D3-D30 streak holders

Instrument opt-out rate + streak maintenance. Frequency cap hard-enforced. Decision gate: opt-out rate flat or declining and D7 retention flat or improving → proceed.

PHASE 2

Days 15-30 · 25% rollout

Activate loss-framing A/B test. Lesson-completion suppression ships. Passive-aggressive notification retired for all users in cohort.

PHASE 3

Days 31-60 · 100% rollout

Lock copy variation pool with content team sign-off. Evaluate P2 items. D30 retention measurement window opens.